

Deepak Vaid

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Personal

Citizen of India

Born on March 30, 1979

Education

Ph.D. Physics, Pennsylvania State University, USA 2003–2012

B.S. Physics, Magna Cum Laude, Missouri University of Science and Technology, USA¹, 2000–2003

Employment

Visiting Associate, Inter-University Centre For Astronomy And Astrophysics (IUCAA), Pune, India, Aug 2017 - May 2020

Assistant Professor, National Institute of Technology Karnataka (NITK), July 2013–Present

Teaching Assistant, Pennsylvania State University, 2003–2007

Physics Tutor, University of Missouri at Rolla, 2000–2002

Research Highlights

A list of my papers can be found at <https://inspirehep.net/search?p=exactauthor%3AD.Vaid.1>

My work is on the interface of many body physics and quantum gravity.

Ongoing Projects

1. I am working on a model which shows the emergence of a cosmological arrow of time from symmetry breaking in spin-network states of Loop Quantum Gravity.
2. I am exploring the application of deep-learning and neural networks to understanding phases of many-body systems. For this I am employing the software packages known as *Theano* and *PyTorch*.

¹[previously known as](#) “University of Missouri at Rolla” (UMR)

3. With my PhD students I am exploring the connection between bulk thermodynamics of charged black holes in anti-deSitter spacetimes and the physics of the horizons of these black holes as described by the Damour-Navier-Stokes' equations.
4. I am working to obtain a better understanding of dynamics in Loop Quantum Gravity via the action of the Hamiltonian constraint operator on spin-network states.

Completed Projects

1. I have co-authored a book on Loop Quantum Gravity, which has recently been published by Springer (ref: [LQG for the Bewildered](#), also available online at [arXiv:1402.3586](#)).
2. In [arXiv:1711.05693](#) I have argued that the Nambu-Goto action of the bosonic string theory should be viewed as arising from the expectation value of the LQG area operator acting on an ensemble of spin-networks. This work represents one of the first pieces of evidence for a direct relationship between String Theory and Loop Quantum Gravity.
3. In [arXiv:1607.00955](#) I have shown that Kepler's Second Law can be seen as a direct consequence of applying the *thermal time hypothesis* to the motion of planetary bodies.
4. In [arXiv:1312.7119](#) I proposed a relationship between Zhang's $SO(5)$ theory (which unifies superconductivity and anti-ferromagnetism) with gravity arising from the so-called BF formulation. I've conjectured that different spacetimes can be thought of as different phases of an underlying condensed matter theory.
5. In [arXiv:1208.3335](#) I pointed out the relationship between the large diffeomorphisms of black hole horizons and the physics of the quantum hall effect. This relationship was also elaborated on by other authors in [arXiv:1402.2274](#). In the same paper I proposed that the dynamics of a black hole horizon can be modeled using the Hubbard model. This idea has been rediscovered recent in work by other authors [arXiv:1703.01156](#)
6. In [arXiv:1307.0096](#) I proposed a relationship between elementary particles and quantum computational gates. This model provides a concrete basis for the so-called computational universe hypothesis.
7. The origin of gauge fields corresponding to the strong, weak and electromagnetic forces remains a mystery. Several proposals exist:
 - (a) Grand Unified Theories (GUTs) based on symmetry breaking of large gauge groups ($SO(5)$ or $SO(10)$),
 - (b) String models where Standard Model gauge fields and various scalar fields arise due to compactification of the 10D stringy spacetime,
 - (c) As additional degrees of freedom which can be attached to spin-networks in the LQG approach to quantum gravity.

In [arXiv:1309.0652](#) I have proposed that, instead, that so-called "fundamental" gauge fields might instead correspond to emergent degrees of freedom required in order to explain the interaction between defects in spin-networks which provide the microscopic substructure for spacetime in LQG.

Publications/Preprints

1. *Arrow of Time from Spontaneous Symmetry Breaking in Quantum Gravity*, under preparation, 2018
2. Connecting Loop Quantum Gravity and String Theory via Quantum Geometry, 2017, [arXiv:1711.05693](#)

3. LQG for the Bewildered, (with Sundance Bilson-Thompson), 2014, [arXiv:1402.3586](#)
Springer Nature, 2017, doi:[10.1007/978-3-319-43184-0](#)
4. Thermal Time and Kepler's Second Law, 2016, [arXiv:1607.00955](#)
5. Quantum Gravity for Dummies, 2014, [arXiv:1402.2757](#)
6. Superconducting and Anti-Ferromagnetic Phases of Spacetime, 2013, [arXiv:1312.7119](#)
Advances in High Energy Physics, Volume 2017, Article ID 7935185, doi:[10.1155/2017/7935185](#)
7. Non-abelian Gauge Fields from Defects in Spin-Networks, 2013, [arXiv:1309.0652](#)
8. Elementary Particles as Gates for Universal Quantum Computation, 2013, [arXiv:1307.0096](#)
9. Quantum Hall Effect and Black Hole Entropy in Loop Quantum Gravity, 2012, [arXiv:1208.3335](#)
10. Embedding the Bilson-Thompson model in an LQG-like framework, 2010, [arXiv:1002.1462](#)
11. Local Void vs Dark Energy: Confrontation with WMAP and Type Ia Supernovae (with S. Alexander, T. Biswas and A. Notari), 2008, [arXiv:0712.0370](#)
Journal of Cosmology and Astroparticle Physics 0909:025,2009, doi:[10.1088/1475-7516/2009/09/025](#)
12. A fine tuning free resolution of the cosmological constant problem (with S. Alexander), 2007, [arXiv:hep-th/0702064](#)
13. Gravity Induced Chiral Condensate Formation and the Cosmological Constant (with. S. Alexander), 2006, [arXiv:hep-th/0609066](#)

Talks, Seminars, Research Visits

1. *Connecting LQG and String Theory: From Quantum Geometry to the Nambu-Goto Action*, seminar at the Physics Department, Birla Institute of Technology Pilani at Hyderabad ([BITS-Hyderabad](#)), Aug 14, 2018
2. *Arrow of Time from Spontaneous Symmetry Breaking in Quantum Gravity*, Poster presentation at workshop on "Optimising, Renormalising, Evolving and Quantising Tensor Networks" ([EVONET18](#)) held at Max Planck Institute for the Physics of Complex Systems, Dresden, Germany, Jun 18-22, 2018
3. *Connecting LQG and String Theory: From Quantum Geometry to the Nambu-Goto Action*, talk at 6th Tux Workshop on Quantum Gravity, Tux, Austria, Feb 19-23, 2018
4. Research Visit under Associateship Programme, Inter-University Centre For Astronomy And Astrophysics (IUCAA), Pune, India, Dec 29, 2017 - Jan 16, 2018
5. *Differential Equations in Physics: From Newton to Einstein*, invited talk at National Workshop on Topics In Partial Differential Equations, NITK Surathkal, July 28, 2017
6. *Phases of Spacetime: Gravity as a Condensate of Gauge Fields*, Talk in "Quantum Gravity" Parallel Session, IAGRG-29, May 18 - May 20, 2017, IIT Guwahati
7. Research interaction with Prof. Romesh Kaul and Prof. Ghanshyam Date at IMSc Chennai from Apr 5-11, 2017
8. *Phases of Spacetime: Gravity as a Condensate of Gauge Fields*, Poster presentation and "Promotional Talk for Young Scientist", DAE-BRNS Symposium on High Energy Physics at Delhi University, December 12-16, 2016 (ref: [arXiv:1312.7119](#))

Last updated: November 10, 2018

9. Research Interaction undertaken under TEQIP-II program of Govt of India with Prof Pablo Arrighi at ENS Lyon, France & LIF Marseille, France, June 9-20, 2016
10. *Phases of Spacetime: Gravity as a Condensate of Gauge Fields*, Poster presentation, International Conference on Gravitation and Cosmology (ICGC 2015) at IISER Mohali, India, December 14-18, 2015 (ref: [arXiv:1312.7119](#))
11. *Phases of Spacetime: Gravity as a Condensate of Gauge Fields*, Seminar, School of Physics, University of Hyderabad, Hyderabad, India, September 7, 2015 (ref: [arXiv:1312.7119](#))
12. *The ABC of Quantum Gravity*, National Institute of Technology Karnataka (NITK), INSPIRE Talk, Surathkal, India, December 16, 2013 (ref: [arXiv:1402.2757](#))
13. *Loop Quantum Gravity for the Bewildered*, Colloquim, Center for Theoretical Physics, Jamia Milia Islamia, New Delhi, India, June 14, 2012 (ref: [arXiv:1402.3586](#))
14. *Loop Quantum Gravity for the Bewildered*, Three Seminars given at Physics Department, University of Adelaide, Adelaide, Australia, August 19, 22 and 29, 2011 (ref: [arXiv:1402.3586](#))
15. *Elementary Particles as Gates for Universal Quantum Computation*, Colloquim at Center for High Energy Physics, Indian Institute of Science, Bangalore, India, April 21, 2010 (ref: [arXiv:1307.0096](#))
16. *Elementary Particles as Gates for Universal Quantum Computation*, Seminar at Mehta Research Institute, Allahabad, India, April 7, 2010 (ref: [arXiv:1307.0096](#))
17. October 2009, [Visiting Researcher](#) at Perimeter Institute for Theoretical Physics, Waterloo, Canada

Computational Skills

Languages: Python, Jupyter (previously IPython Notebook), C++, L^AT_EX, SAGEMath, Mathematica, Git, HTML

Platforms: Apple MacOS, Linux, Microsoft Windows

References

[Stephon Alexander](#), Professor of Physics, Brown University, RI, USA, stephon_alexander@brown.edu

[Martin Bojowald](#), Associate Professor of Physics, 304A Whitmore Lab, Pennsylvania State University, PA, USA, +1 814 865 3502, bojowald@gravity.psu.edu

Sundance Bilson-Thompson, Visiting Researcher, Department of Physics, University of Adelaide, Adelaide, Australia, sundancebt@gmail.com

G. Umesh, Professor (Retired), Department of Physics, National Institute of Technology Karnataka, Surathkal - 575025, Karnataka, India, +91-9901184376, +91-824-2473276, umesh52@gmail.com

Teaching Activities

Assistant Professor, National Institute of Technology Karnataka (NITK):

Modern Physics, Introductory Modern Physics Course for B. Tech 1st year students, Aug 2013 – Present

Instructor, Aug 2013 – Present (Ten semesters)

Course Co-ordinator, Aug 2014 – May 2015

Instructor, *Mathematical Physics*, M.Sc. Physics 1st year, Aug 2013 – Dec 2013

Instructor, *General Relativity*, M.Sc. Physics Elective, Jan 2014 – May 2014

Instructor, *Nuclear Physics*, M.Sc. Physics 2nd year, Aug 2014 – Dec 2014, Aug 2016 – Dec 2016

Instructor, *Quantum Hall Effect*, as part of *Material Synthesis & Characterization*, Oct 2015

Instructor, *Superconductivity*, as part of *Condensed Matter Physics, Part 2*, Oct 2016 – Nov 2016

Teaching Assistant, Pennsylvania State University:

Conducted Labs for Introductory Physics Courses, Aug 2003 - Dec 2007 (Seven Semesters + Four Summer Sessions)

Teaching Experience (Theory)				
Course Code	Course Title	PG/UG	Period	Credit Hours
PH110	Physics (*)	UG	Jul-Dec, 2013	4
PH701	Mathematical Physics	PG	Jul-Dec, 2013	4
PH110	Physics	UG	Jan-May, 2014	4
PH876	General Relativity	PG	Jan-May, 2014	3
PH110	Physics	UG	Jul-Dec, 2014	4
PH862	Nuclear Physics	PG	Jul-Dec, 2014	2 (**)
PH110	Physics	UG	Jan-May, 2015	4
PH110	Physics	UG	Jul-Dec, 2015	4
PH914	Applied Quantum Mechanics	PG	Jul-Dec, 2015	3
PH879	Physics of Low Dimensional Systems	PG	Jul-Dec, 2015	1 (***)
PH110	Physics	UG	Jan-May, 2016	4
PH110	Physics	UG	Jul-Dec, 2016	4
PH802	Nuclear Physics	PG	Jul-Dec, 2016	2 (**)
PH801	Condensed Matter Physics, Part II	UG	Jul-Dec, 2016	1 (****)
PH110	Physics	UG	Jan-May, 2017	4
PH753	Statistical and Thermal Physics	PG	Jan-May, 2017	2 (**)
PH876	General Relativity	PG	Jul-Dec, 2017	3

Table 1: *Legend:* (*) 1st year introductory modern physics course for B.Tech students; (**) 4 credit hour course shared with Prof H. S. Nagaraja; (***) Taught “Hall Effect” module as part of PH879 (overall course instructor Prof Partha P. Das); (****) Taught “Superconductivity” module as part of PH801 (overall course instructor Prof Kasturi Banger)

Research Supervision

Ph.d. Students Supervision

Name	Title	From	To
Ahmed Rizwan C. L.	Geometric Thermodynamics and Holographic Superconductivity	Dec, 2013	Present
Rajani K. V.	Bulk vs Boundary Dynamics of Charged Anti-deSitter Black Holes	Dec, 2014	Present
Ananthram Bhat	Using Deep Learning to Identify Phases of Many-Body Systems	Aug 2017	Present

M.Sc. Students Supervision

Name	Title	From	To
Arya Antherjanam	Green's Functions and SYK Model of Holography	Aug, 2017	May 2018
Nrusingha Prasad	Weak Localization using Random Potentials	Aug, 2017	May 2018
Merin Joseph (*)	Chaotic Dynamics in Complex Systems	Aug, 2016	May, 2017
Shwetha B.	Relativistic Velocity Addition Law Without the Second Postulate	Aug, 2015	May 2016
Sreedevi Varma N. (**)	Alcubierre's Drive: A Study Using ADM Formalism and Energy Conditions	Aug, 2015	May, 2016
Priyamedha S K R (***)	Electrodynamics of Cavity Resonators	Aug, 2014	May, 2015
Sarath E.	Perturbation Theory and Relativistic Corrections	Aug, 2014	May, 2015

Table 2: *Legend:* (*) Pursuing PhD in soft condensed matter physics at the University of Leeds, UK, August 2018 onwards; (**) Pursuing PhD in applying machine learning to cosmology and astroparticle physics at King's College London, UK, Fall 2016 onwards; (***) Pursuing PhD in Physics at IIT Mandi, India, Fall 2016 onwards.

Research Supervision (*contd.*)

Undergraduate Research Internships			
Name	Home Institute	Project Title	Period
Rajashree Kamath	Vellore Institute of Technology, Chennai, B.Tech (2018)	Exploring Decoherence with QuTiP (Quantum Toolbox in Python)	Oct 2018 – present
Shalmali Sudhindra	Alva's College, Moodbidri	Non Equilibrium Greens function and Electrical conductivity in small scale systems	Jun - Jul, 2017
Karthik Suresh	Central University of Karnataka, Gulbarga	Conductance in Mesoscopic Systems via Non-equilibrium Green's Functions	Jun - Jul, 2017
Neel Kumar	Vellore Institute of Technology, Chennai	Simulating Planetary Orbital Motion	Jun - Jul, 2016
Vinayaka Garg	Vellore Institute of Technology, Chennai	1D Ising Model	Dec 2015